WORKSHEET

Using the binomial probability distribution

1 Twenty-one per cent of people experience some kind of learning difficulty while at school. In a class of 30 students, what is the probability that:

a four experience learning difficulties?

b seven experience learning difficulties?

c zero experience learning difficulties?

d 12 experience learning difficulties?

e 25 have no learning difficulties?

f no more than two have learning difficulties?

2 A multiple-choice test has four possible answers for each of 10 questions. If a student has not studied and randomly guesses the answers, what is the probability that they get:

a five correct answers?

b zero correct answers?

c one correct answer?

d six incorrect answers?

e fewer than three answers correct?

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3 In a clinical trial, 60% of participants receive a trial drug while the rest are given a placebo. If 50 of the participants are collected in one room, what is the probability that:

a 30 are receiving the trial drug?

b 50 are receiving the trial drug?

c 25 are receiving the placebo?

d 10 are receiving the placebo?

4 Given that it is equally likely that a child will be born male or female, if a family has six children what is the probability that there are:

a two girls?

b five girls?

c zero girls?

d three boys?

e more than three girls?

f fewer than two boys?

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Answers

1 **a** ≈ 0.1161

b ≈ 0.1621

c ≈ 0.0008

d ≈ 0.0091

e ≈ 0.1605

f ≈ 0.0337

2 **a** ≈ 0.0584

b ≈ 0.0563

c ≈ 0.1877

d ≈ 0.1460

e ≈ 0.5256

3 **a** ≈ 0.1146

b $\approx 8.083 \times 10^{-12}$

c ≈ 0.0405

d ≈ 0.0014

4 **a** ≈ 0.2344

b ≈ 0.0938

c ≈ 0.0156

d 0.3125

e ≈ 0.3438

f ≈ 0.1094

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